Performance Limitations and Optimizations of Linear Driver Optics for 200G/Lane and beyond

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Abstract: We studied performance limitations and optimizations using digital equalization and modulation bandwidth in linear-driver-optics for 200G/lane and beyond. The results show linear pluggable modules can support 30dB bump-to-bump loss for 200G/lane using an enhanced CTLE. (e-mail: jzhou@hisensebroadband.com). © 2024 The Author(s)

1. Introduction

Linear-drive-optics (LDO) in optical transceiver modules, which eliminates digital signal processing (DSP) in the module, aligns well with the requirements of AI/ML applications such as low power, low latency and low cost [1-3]. There are two types of LDO form factors: one is Linear-drive Pluggable Optics (LPO), and the other is linear Co-Package Optics (CPO). Although linear CPO has low RF connection loss in switch, LPO modules show promise to be adapted to links with existing DSP based pluggable modules. 100G/lane LPO was demonstrated for commercial use, while 200G/lane in LPO still raises concerns due to high RF connection loss in switch. The simulations show the feasibility of retiming the transmitter (Tx) and linear receiver (Rx) with half DSP [2,3]. However, LPO is still attractive to eliminate DSP, especially having a simple architecture with an absolute low latency.

In LPO applications, electrical signals from Serdes are connected to the linear driver and modulator in the module via RF connection. In a 51.2T switch, the host ASIC to optical module electronic integrated circuit (EIC) can have ball-to-ball loss up to 15dB at Nyquist frequency of 26GHz; however, for 200G/lane LPO modules in a 128-radix cable-routed switch, the worst-case bump-to-bump loss is estimated to be 30dB at Nyquist frequency of 56GHz [2]. E. S. Chou, et al, show that 200G/lane LPO requires bump-to-bump loss of <23dB to achieve 3.5dB ER and 3.9dB TDECQ [2]. This indicates that improvement is essential for 200G/lane in LPO applications. The system performance is limited by multiple design factors, including bump-to-bump loss, bandwidth of linear optics, and digital equalization. The digital equalization can compensate for a low modulation bandwidth and/or high RF connection loss; however, the performance is strongly dependent on the equalization bandwidth [5].

In this paper, we studied the impact of digital equalization and modulation bandwidth on system performance for 200G/lane applications. We propose an enhanced CTLE at transmitter (Tx) to overcome the performance limitations. Our simulation data shows that the performance can be achieved for 200G/lane LPO with 31dB bump-to-bump loss by design optimization using an enhanced CTLE of 5.6dB at Tx and TFLN MZM with 3dB BW of 90GHz. The performance includes transmitter dispersion and eye closure quaternary (TDECQ) of <2.5dB, extinction ratio (ER) of >3dB, receiver (Rx) sensitivity of -4dBm, and BER floor below 1E-6. Our analysis indicates that an enhanced CTLE can be applied to 400G/lane to improve performance for LDO, especially in linear CPO applications with low bump-to-bump loss.

2. Device and Model Parameters

We studied performance limitations in LPO for 200G/lane applications with various bump-to-bump loss, bandwidth of linear optics, and digital equalization. To minimize the nonlinear impact, we used Silicon Photonics (SiP) and Thin-film Lithium Niobate (TFLN) based Mach-Zehnder modulator (MZM) and linear driver, which are designed for 200G/lane applications. Fig. 1 a) shows electro-optic (EO) small signal frequency response of LPO Tx with SiP and TFLN MZMs. The EO response is from the driver input to the Tx optical output including driver, MZM, and their RF interconnect. TFLN MZM design shows 15dB peaking at 50GHz with 3dB BW of 84GHz, while SiP MZM design has 9dB peaking at 50GHz with 3dB BW of 76GHz. For performance simulation with 200G/lane in LPO, we used bump-to-bump losses in a range from 16dB to 31dB at Nyquist frequency at 56GHz with same frequency response in reference [4]. Fig. 1 b) shows frequency response of bump-to-bump loss. To compensate for a high bump-to-bump RF loss, we introduce an enhanced CTLE at Tx. In simulation, we use CTLE with frequency response defined in CEI-224G-LR-PAM4 LR Interface [6]. Fig. 1 c) shows frequency response of CTLE at Tx, with peaking gain of 5.6dB and 11.3dB used in the optimized designs. In an actual implementation, we can design an active CTLE which can be integrated into the linear driver [7].

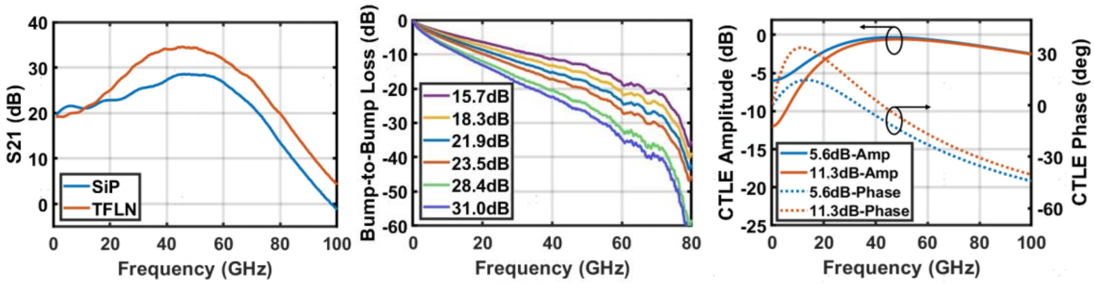

Fig.1 a) Small signal response of LPO Tx from driver input to Tx optical output with SiP and TFLN MZMs (left), b) frequency response of bump-to-bump loss (mid), and c) frequency response of CTLE at Tx (right).

The model for system performance simulation is built with VPI, a commercial simulation software. Fig. 2 shows a function diagram of a model for system performance simulation. Table 1 summarizes the model parameters.

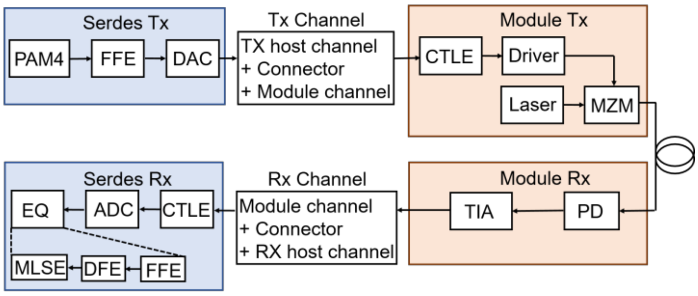

Fig. 2 Function diagram of a model for performance simulations.

Table 1 Model Parameters

Location	Components	Parameters
Serdes Tx	EQ and DAC	7-tap FFE (3 pre, 3 post), 56GHz DAC 3dB BW
Module Tx	MZM	TFLN: 90GHz 3dB BW, 4V V_π; SiP: 50GHz 3dB BW, 6.5V V_π
	CW laser	1MHz linewidth, -145dB/Hz RIN, 1310nm wavelength
	Driver+MZM	TFLN: 84GHz BW, 15dB peaking at 50GHz, 2Vppd driver output swing
		SiP: 76GHz BW, 9dB peaking at 50GHz, 3Vppd driver output swing
	CTLE	6dB CTLE gain
Module Rx	PD+TIA	0.7A/W PD response, 56GHz BW, 18 pA/sqrt(Hz) IRN
Serdes Rx	CTLE/EQ	13dB CTLE gain, 38-tap FFE, 2-tap DFE, MLSE
RF channel	TX/Rx channel	15-31dB bump-to-bump loss at 56GHz
Optical link	Fiber	500m

RIN: relative intensity noise, IRN: input referred noise, CTLE: continuous time linear equalizer, EQ: equalizer, FFE: feed-foward equalizer, DFE: decision-feedback equalizer, MLSE: maximum likelihood sequence estimation

3. Results and Discussions

We studied Tx and Rx performance with model parameters summarized in Table 1. The simulation is carried out with 112Gbaud PAM4. We use a pseudorandom binary sequence quaternary (PRBSQ) for Hamming (128,120) code with a forward error correction (FEC) threshold of 4.85×10^{-3} [2]. TDECQ and ER are measured to characterize Tx performance and BER vs Rx OMA is measured to study system performance. Rx OMA sensitivity is defined at a FEC threshold of 4.85×10^{-3}. Tx eye is optimized with 7 FFE taps. Driver output swing is maintained at 2Vppd for TFLN MZM with maximum gain of 18.5dB and at 3Vppd for SiP MZM with maximum gain of 28dB for SerDes output swing of 1200mV [6]. An enhanced CTLE is introduced to improve TDECQ and Rx BER performance. We studied the impact of digital equalization with 5bit and 7bit resolutions and compared them with analog FFE based equalization. We select 7bit resolution for the simulation in this paper as it shows slightly better performance than 5bit, while analog equalization shows much better performance than digital equalization, under the worst bump-to-bump loss of 31dB.

Fig. 3 a) shows TDECQ and ER vs bump-to-bump loss with SiP and TFLN MZMs. Fig. 3 b) shows BER vs Rx OMA. We study the performance under the worst bump-to-bump loss of 31dB. The results show that TDECQ is 6.5dB without CTLE and is improved to 3.3dB with an enhanced CTLE of 5.6dB for SiP MZM, while an analog equalization can achieve TDECQ of 1.9dB. For TFLN MZM, TDECQ is 2.7dB without CTLE and 2.1dB with the CTLE of 5.6dB whose eye is inserted. The ER is 3.1dB for TFLN MZM with driver output swing of 2Vppd, while the ER is 2.9dB for SiP with driver output swing of 3Vppd. For Rx performance with TFLN MZM, the results show that BER floor is at 1E-5 and can be improved to be less than 1E-6 with the enhanced CTLE of 5.6dB at Tx, while Rx OMA sensitivities are -4.5dBm and -4.1dBm with and without the CTLE, respectively. However, the BER floor is above 1E-5 for SiP MZM with the CTLE. The Rx OMA sensitivity is -2.4dBm and -2.2dBm for the CTLE of 5.6dB and 11.3dB, respectively. When bump-to-bump loss is reduced to 22dB, the BER floor is below 1E-6 for both TFLN and SiP MZMs, although TFLN MZM shows 1.5dB better Rx sensitivity than SiP MZM.

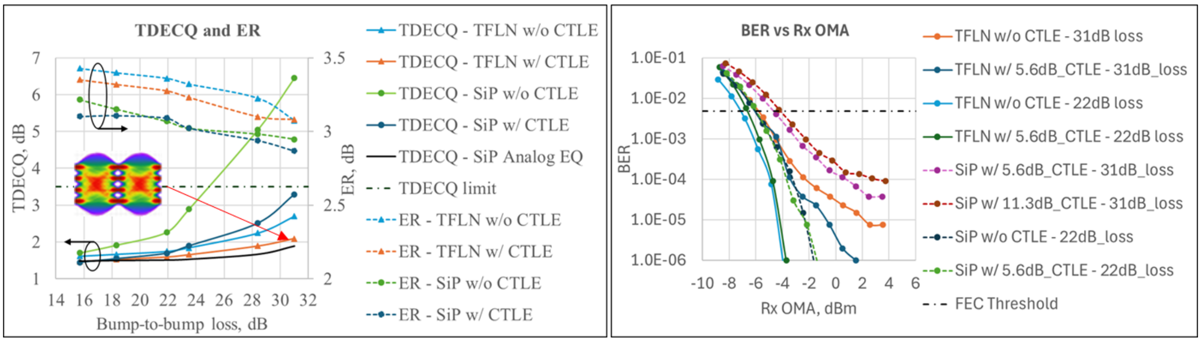

Fig. 3 a) TDECQ and ER vs bump-to-bump loss (left), and b) BER vs Rx OMA (right).

The results with 200G/lane show that TFLN with high BW can support the worst bump-to-bump loss of 31dB by compensation with an enhanced CTLE at Tx to meet the performance requirements. The BER measurement below 1E-6 is limited by the model and will be measured by an improved model. For SiP MZM with the CTLE, BER floor stays above 1E-5, although TDECQ below 3.5dB is achieved. It is worth noting that performance will be limited due to high peak-to-average-power-ratio (PAPR) when only digital equalization is used to compensate low modulator bandwidth and/or high RF connection loss [5]. Therefore, the performance can be optimized by trade-off between digital equalization and analog equalization (CTLE). For LDO with 400G/lane, RF connection loss is expected to double compared with 200G/lane if the same RF connection is used. The bandwidth of linear driver and modulator will also be limited. Like 200G per lane, an enhanced CTLE at Tx can be applied to 400G/lane to improve the performance. It is foreseen that 400G/lane may be more challenging to be used in LPO; but remains promising in LDO, especially in linear CPO applications.

4. Conclusions

We studied the impact of digital equalization and bandwidth of linear optics with various bump-to-bump loss and modulators for 200G/lane modules in LPO applications. Our results show that an enhanced CTLE at Tx can overcome performance limitations due to high bump-to-bump loss. The performance, with TDECQ of < 3dB, ER of > 3dB, Rx BER floor of < 1E-6, and Rx OMA sensitivity of -4.5dBm, is achieved under 31dB bump-to-bump loss by design optimization with an enhanced CTLE of 5.6dB and 90GHz BW TFLN MZM. The BER floor stays at 1E-4 with 50GHz BW SiP MZM under 31dB bump-to-bump loss although the enhanced CTLE is used. The same technique to optimize the performance with trade-off between digital equalization and analog equalization with an enhanced CTLE can be used in LDO, especially in linear CPO, for 400G/lane applications.

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